Names? format? -5

76/95

CASE EXAM 1 - REPORT OF GROUP 7

CONTENT CREATOR'S PERFECT TIMING ENGAGEMENT

justified

In this case study we are investigating whether a content creator should publish her videos on Instagram or Tiktok. Furthermore, we are analyzing the number of views based on Saturday and Wednesday which also depends on the platform. Comparison will include statistical analysis which are statistical plots and descriptive statistics. In our first part, we will focus on mean, variability, and skewness of two platforms which are Instagram and Tiktok. In the second part, we will look at the number of views on different days and interpret it by statistical analysis and determine which day is more logical to post. In the third part we will use estimators and analyze the results of the decision-making process for posting content. Lastly, we will wrap our statistical results and decide which day and platform Cemre should post.

1) Firstly we found statistical data such as mean, variability and skewness. The sample mean is formulated as summing up all the views in the platform and dividing by sample size. According to our calculations the mean of the Tiktok is 10084.7917 and for Instagram is 9423.8130 (table 1.1 & table 1.2) . This shows that for our sample data Tiktok is more preferable than Instagram. From the box plot (figure 2.2) which includes first, second and third quartiles the variability of Tiktok is higher than the Instagram. Also we can assure that with Standard deviation. Both platforms are rightly skewed, but since Tiktok's skewness is lower than Instagram's, the box plot for Tiktok is more likely to be symmetric. The experiment depends on the day and the platform, it does not consturct random sample, but when we divide into weeks?, we see that it is actually random, as it is seen in the time series plot (figure 2.4). It is likely to close normal sample distribution which is also daily dependent. Moreover, the distribution behavior can be seen at normal distribution plots (figure 2.3) and Tiktok is closer to the normal distribution. For the outliers, it is seen that Tiktok's outlier (figure 2.2) is closer to the third quartile therefore it is not required to exclude it. On the other hand, Instagram outlier is extremely deviating from the third quartile. So we are excluded from the data set.

   no need ? -3 ? why? ? not good reasons

2) Since we assume μ_S and μ_W represent the true average views for all Saturday and Wednesday posts, respectively; the mean of Saturday which is 11214.4380 is larger than the mean of Wednesday which is 8294.166 (table 1.3 & 1.4). We assume it is a random sample. To find samples mean? we add all observations and divide them by the observation number. Since we assume it is close to normal distribution we can conclude that all Xi have the same

   no need?

two samples?

mean. Therefore the formula shows that the expectation of this random sample mean is equivalent to μ. Since their expected value and real value is the same their differences are zero. This shows that they are unbiased estimators. We used two sided intervals rather than one sided bounded.

? Since we have sample data that does not go to infinity, we have two limits. To compare saturday and wednesday we need to find confidence intervals for the difference of two population means. Since both of the intervals do not contain zero, we conclude that they have a meaningful difference between Saturday and Wednesday. We conclude that posting on Saturday is more logical since the subtraction from wednesday to saturday is a negative value. -4544.956~~685084308~~, -1295.58~~60222854124~~ with 95% of confidence interval.

which interval?

-3

around $\mu_w - \mu_s$?

3) From the time series plot (figure 2.4) we have concluded that Instagram and Tiktok are not time series dependent (considering the total of weeks not the differences between wednesday and saturday particularly) and can be assumed to be random. Since we can assume randomness we can use the sample proportions as estimators for the true proportions. We have denoted the proportion of posts that exceed 10000 views on Saturday on Instagram as p1 and found it to be equal to 0.54. We have then denoted the proportion of posts that exceed 10000 views on Saturday on Tiktok as p2 and found it to be equal to 0.67. We can clearly observe that the p2 is higher than p1 however we can not reach any conclusions on the variability of the proportions. Since we have already assumed randomness and also normality in part(a) by looking at the normal probability distribution plot we can use confidence intervals. We have determined the confidence level of the CI of p1 and p2 to be %95 and calculated the relative CI's accordingly. Finding the confidence interval is important because by finding the proportions we could not determine necessary variability which is also crucial. We have found the CI of the p1 as (0.351,0.720) and we have found the CI of p2 to be (0.465,0.820). From our findings we can conclude that estimated p2 is larger than p1 and p2 also has lower variability compared to p1. These findings show that prioritizing Tiktok when posting on Saturday would be the more profitable option for Cemre.

CI on $p_1 - p_2$?

-5

4) From the second question, we observed confidence intervals differences between the days of saturday and wednesday. We concluded that Cemre should post her videos on Saturday to increase her view. Since we found the day as Saturday we compared two platforms based on Saturday. The third question concludes that if Cemre gets more than 10000 views she gains more money based on both platforms. The proportion of Tiktok is higher than

contrast between higher mean & variance

-3

Instagram. To share on Tiktok on Saturday provides more money and interaction to her.

APPENDIX

WENDESDAY (1.1)

Mean	Std Dev	Min	Max	1st Quart	Median	3rd Quart	Skewness
8294.1667	2749.8386	2720.0000	15800.0000	6570.0000	8290.0000	9540.0000	0.6231

Saturday (1.2)

Mean	Std Dev	Min	Max	1st Quart	Median	3rd Quart	Skewness
11214.4380	5041.9131	2190.0000	25920.0000	8520.0000	11790.0000	13565.0000	0.6085

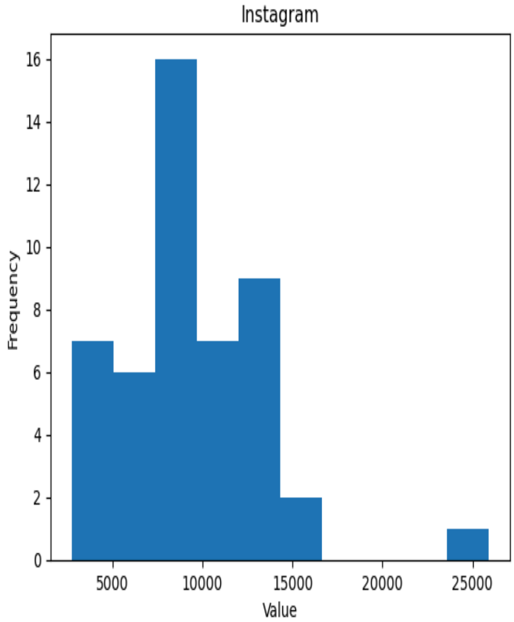
INSTAGRAM (1.3)

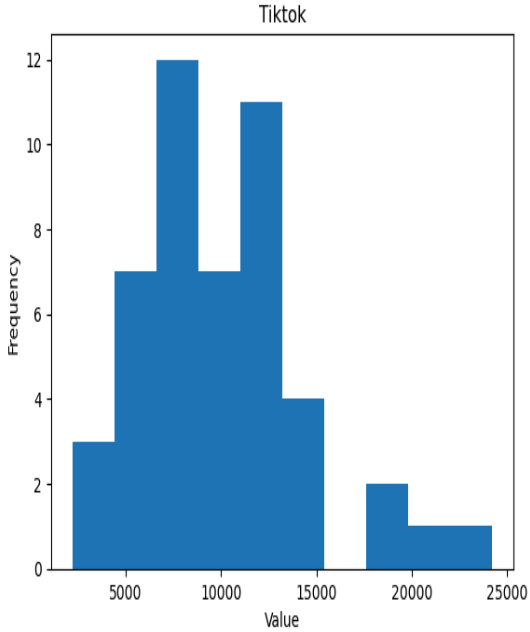
Mean	Std Dev	Min	Max	1st Quart	Median	3rd Quart	Skewness
9423.8130	4091.1797	2720.0000	25920.0000	6840.0000	8815.0000	12017.5000	1.2793

TIKTOK (1.4)

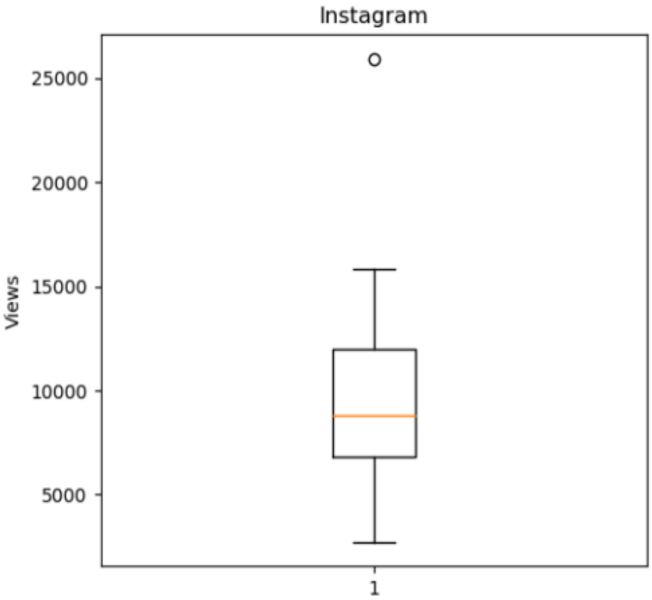
Mean	Std Dev	Min	Max	1st Quart	Median	3rd Quart	Skewness
10084.7917	4514.0193	2190.0000	24230.0000	6805.0000	9415.0000	12527.5000	0.8830

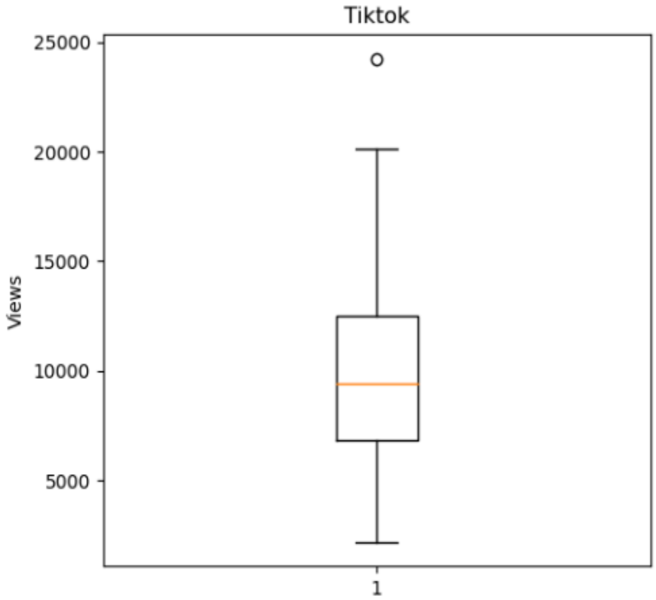
HISTOGRAM (2.1)



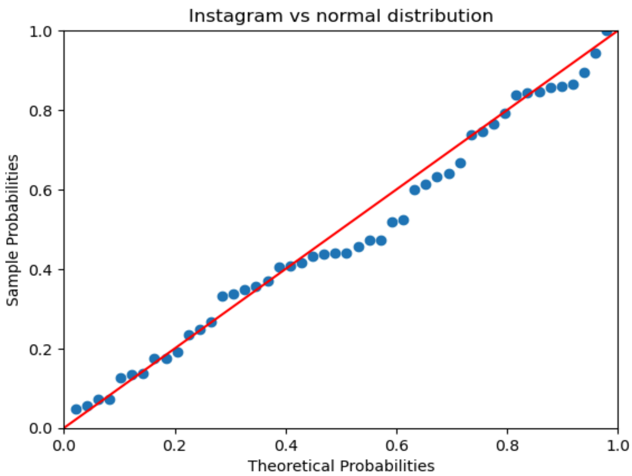


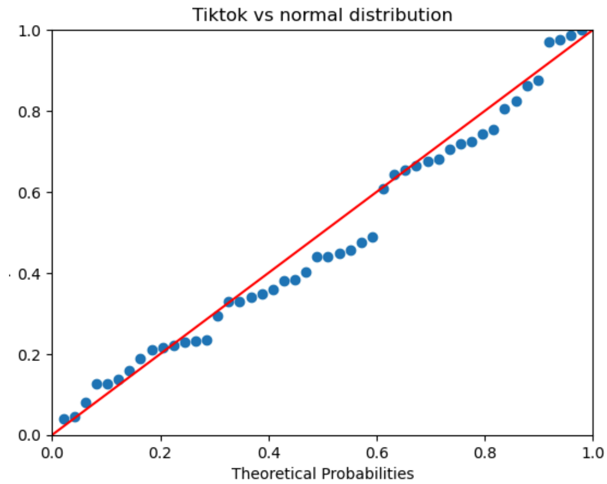
BOT-PLOX (2.2)





NORMAL DISTRIBUTION PLOT (2.3)





TIME SERIES PLOT(2.4)

BLUE - INSTAGRAM
ORANGE-TIKTOK